

Open Science: Publications, Data, Code

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What we mean by Open science:

Goals:

- Make research results available to anyone for free.
- Ensure reproducibility of the results.

Publicly funded research should benefit everybody.

However, these are good practices that can also be applied to private research.

How to achieve this ?

- Make publications open access
- Publish useful datasets
- Publish software
- Share protocols
- Think about it from the start of the project

For publications:

Different ways of publishing:

- Subscriptions are often paid by the University (very expensive)
- Publishing in Open Access often requires the author(s) or the organization to pay
- Some journals are financed in a way that covers Open Access for all (Diamond)

What can you do ?

- Submit to Diamond journals
- Publish the “preprint” on HAL or Arxiv
- Self-archive the “postprint” 6 months after publishing, for example on HAL

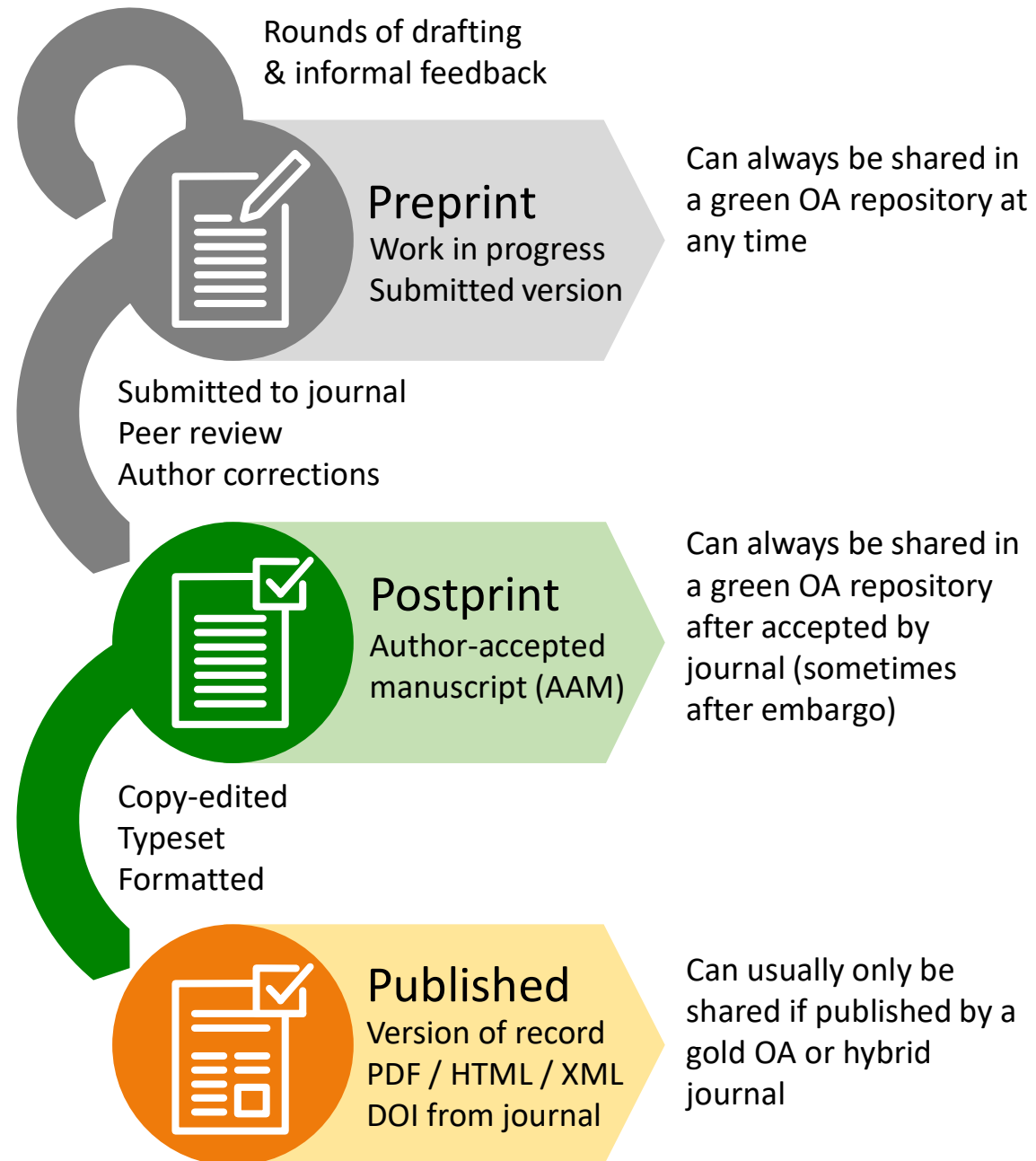
You keep your rights over the publication in any case, so you are free to self-archive your articles (at worst 6 month embargo)



HAL
open science

université
PARIS-SACLAY

Publishing process :



Shafee, T. (2020)
Typical publishing workflow for an academic journal article (preprint, postprint, and published) with open access sharing rights per SHERPA/RoMEO.
Wikimedia Commons, CC BY 4.0 International

For Data:

What is research data ?

*Research data are defined as **factual records** in the form of figures, texts, images and sounds which are used as the **main sources for scientific research** and which the scientific community generally recognises as being **necessary to validate research results**.*

Definition of the OECD

What to do ?

- Generalise publishing data
- Make your data easy to reuse
- Look for pre-existing data before creating new sets
- Justify why some can't be shared

Research data should be
"As open as possible, as closed as necessary"



RÉPUBLIQUE
FRANÇAISE

Liberté
Égalité
Fraternité

recherche.data.gouv.fr

FAIR Data

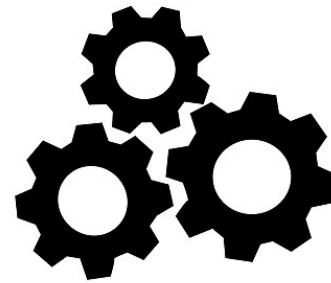
F_{indable}



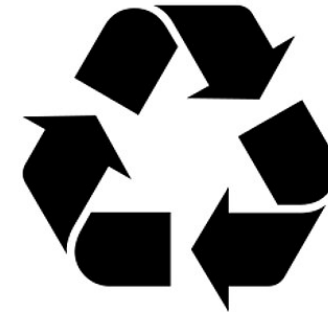
A_{ccessible}



I_{nteroperable}



R_{eusable}



Guidelines for the publication of data

For Software:

Research Software:

- Software used to analyse data
- Software that is itself a research result
- Omnipresent in research
- Impossible to reproduce results without it
- Opening the source code early allow for improvements from other people

What to do ?

- Learn about Open Source methods / use a forge
- Make code available
- Add a Open Source **licence**
- Save it forever with Software Heritage
- Publish it with articles or on HAL

Trying to apply FAIR principles to Software is a very good start

About forges



GitLab

gitlab.dsi.universite-paris-saclay.fr

Your personal installation

The public version gitlab.com

Also possible : GitHub (owned by Microsoft), SourceForge, etc.

A forge allows to work collaboratively on the same project without conflict, as you work on your computer and push changes on the project when it's ready to be incorporated

Software Heritage



Software Heritage

THE GREAT LIBRARY OF SOURCE CODE

Harvests public repositories from different forges to save source code forever, assigning it a permanent identifier called SWHID

Makes sure code will be preserved and identified forever !
New versions are saved in the same place



Can be published on HAL, linked to datasets, articles...

What about AI ?

Emerging questions related to datasets and
their use in AI

How to consider datasets generated by AI ?

What about the copyright for these
datasets ?

How to publish models ?

Contact :

At Université Paris-Saclay :

- science.ouverte@universite-paris-saclay.fr for general inquiries
- donnees-recherche@universite-paris-saclay.fr for data related questions

Open Science Week : 16th to 20th November 2026, focus on data

At Institut Polytechnique de Paris :

- scienceouverte@ip-paris.fr